

# ORANGE BOT – ROBOCUPJUNIOR RESCUE MAZE

## TEAM

### Orange Bots

Country: Portugal

League: RoboCupJunior  
Rescue Maze – Secondary

Objective: Gain experience in robotics and develop an autonomous robot capable of exploring and rescuing victims in a maze environment.

## TEAM MEMBERS

David Mónica – 12 years old  
Electrotechnics Technician Course  
Responsible for hardware assembly and communication configuration between different components.

João Castro – 11 years old  
Programming and Computer Science Course  
Responsible for programming the navigation logic and victim detection.

## EXPERIENCE

This is our first participation in RoboCupJunior competitions.

## CAMERA

Raspberry Pi Camera 3  
(victim detection)

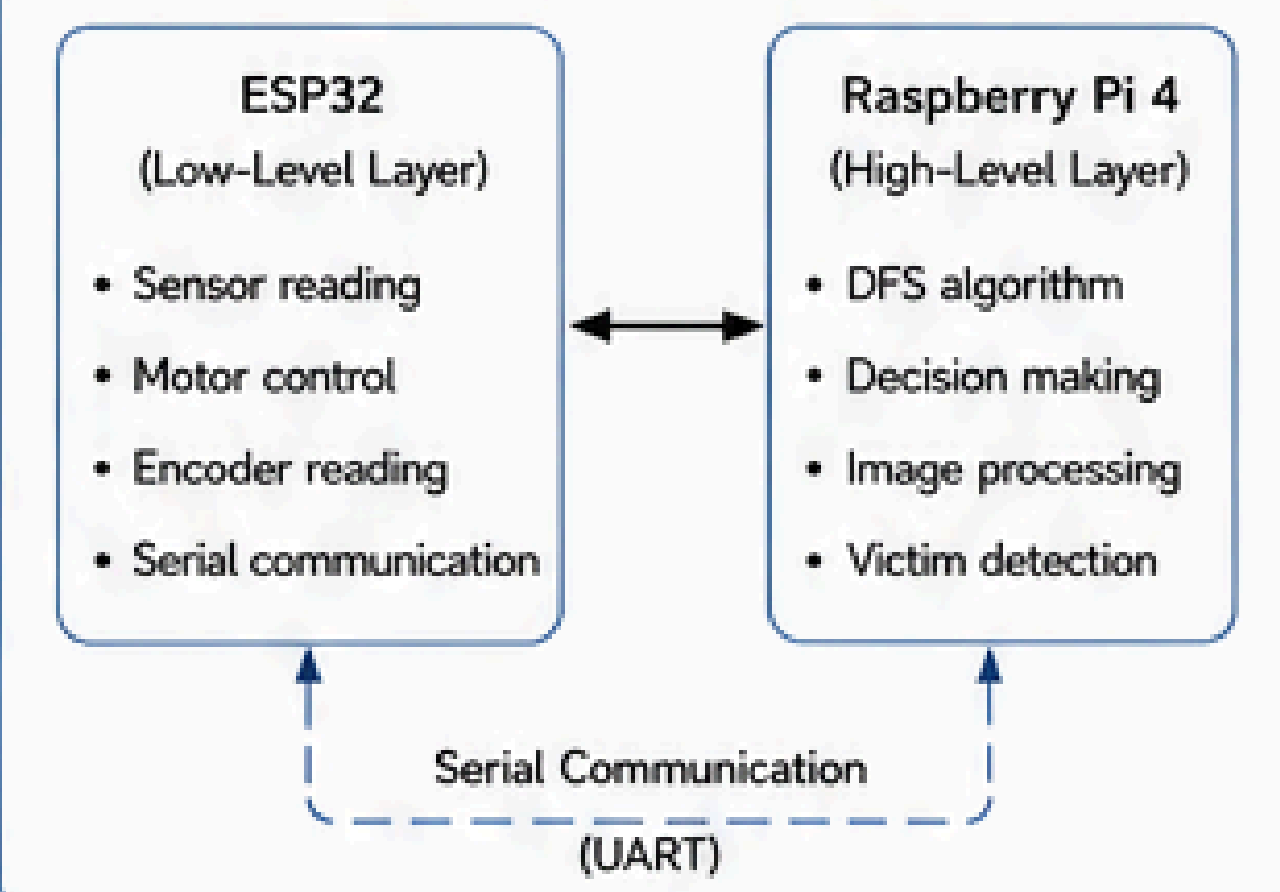
ULTRASONIC  
SENSOR  
Left Side

ULTRASONIC  
SENSOR  
Right Side

ULTRASONIC  
SENSOR  
Front

COLOR SENSOR  
(mark/victim detection)

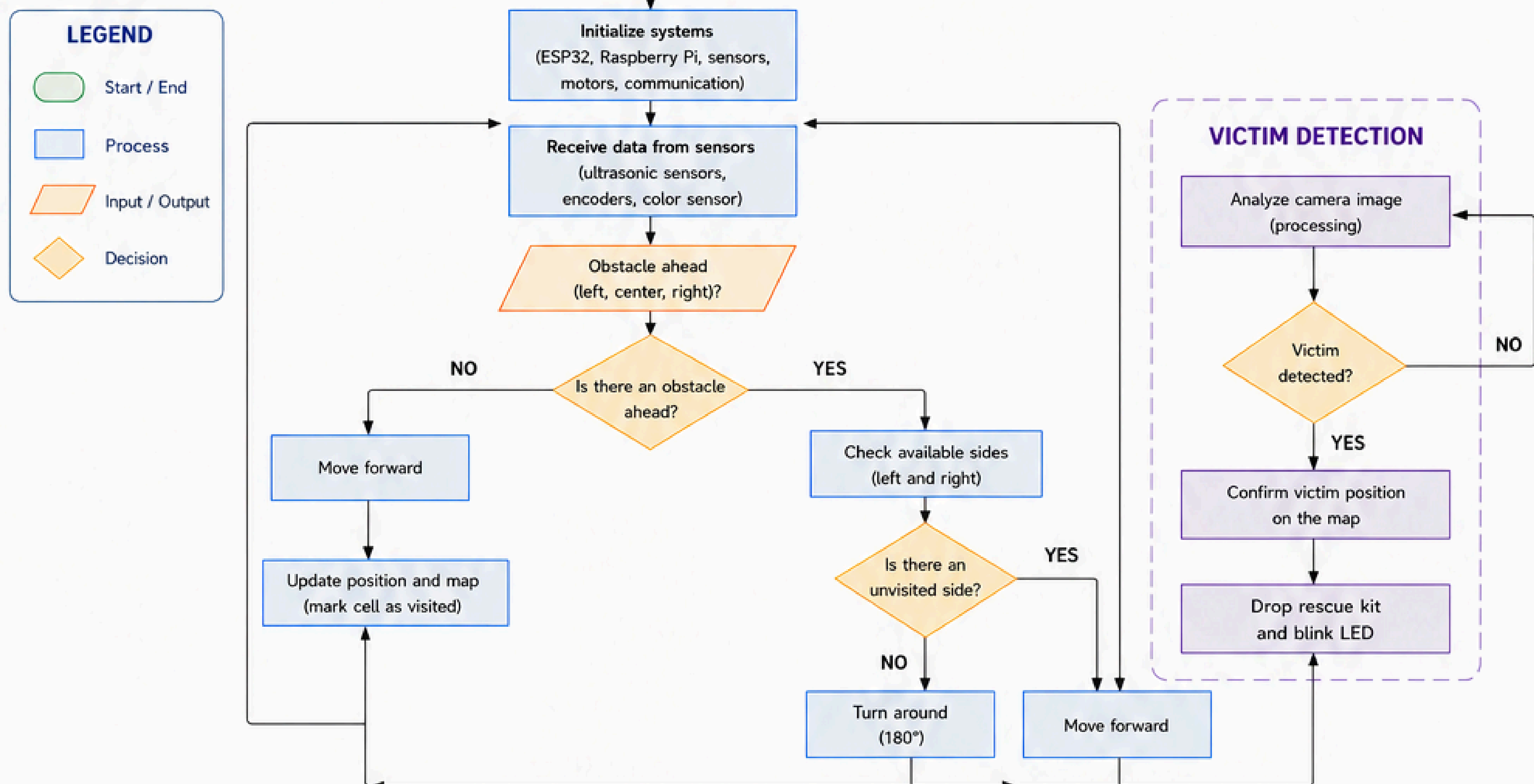
## ARCHITECTURE



## MAIN COMPONENTS

|  |                           |                                                             |
|--|---------------------------|-------------------------------------------------------------|
|  | ESP32                     | Microcontroller responsible for low-level tasks.            |
|  | Raspberry Pi 4            | Computer responsible for high-level logic.                  |
|  | Ultrasonic Sensors (3x)   | HC-SR04 – obstacle detection (front and sides).             |
|  | Color Sensor              | TCS3200 (I2C) – detection of markings/victims on the floor. |
|  | Motors with Encoders (4x) | DC motors with encoders – locomotion and odometry.          |
|  | Camera Module             | Raspberry Pi Camera Module 3 – victim detection.            |
|  | Power Supply              | Li-Po Battery – powers all components.                      |

## FLOWCHART – MAIN LOOP



## MAIN LOOP DESCRIPTION

1. Read data from all sensors and update the robot's state.
2. Check if there is an obstacle ahead.
3. If not, move forward and update the map.
4. If there is an obstacle, check unvisited sides (left or right).
5. If one or both sides are available, turn and move forward.
6. If no sides are available, turn around (backtracking).
7. When a victim is detected, record the position.
8. The loop repeats until the maze is fully explored.

## NOTES

- The algorithm used is DFS (Depth-First Search).
- The maze map is represented as a matrix of cells.
- Each cell can be: unvisited, visited, or with a victim.
- Communication between ESP32 and Raspberry Pi is done via serial (UART).